

THREE DIMENSIONAL GEOMETRY

1+2+3+5 = 11m

If a directed line L passing through origin makes angles α, β, γ with x, y, z axes respectively, then the angles are direction angles and cosine of these angles i.e. $\cos \alpha, \cos \beta, \cos \gamma$ are called DC's of L .

If we reverse the direction of L , the direction angles are replaced by their supplements i.e. $\pi - \alpha, \pi - \beta, \pi - \gamma$ and thus signs of DC's are reversed.

The 3 numbers proportional to DC's of a line are called DR's a, b, c and thus $\frac{a}{l} = \frac{b}{m} = \frac{c}{n} = \lambda$
 $\left[\begin{array}{l} a, b, c \rightarrow \text{DR's} \\ l, m, n \rightarrow \text{DC's} \end{array} \right]$

Identities:

* $l^2 + m^2 + n^2 = 1$

$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$

* If two points $A(x_1, y_1, z_1), B(x_2, y_2, z_2)$ are given on a line L , then DR's of the line are $(x_2 - x_1), (y_2 - y_1), (z_2 - z_1)$

* Collinearity of 3 points:

To prove -

i) $\vec{AB} \parallel \vec{BC}$

ii) B is common point

$\Rightarrow A, B, C$ are collinear

EXERCISE 11.1

Eg 1] If a line makes $\angle$'s $90^\circ, 60^\circ, 30^\circ$ with +ve direction of x, y, z axis. Find its DC's.

Sol] $l = \cos \alpha = \cos 90^\circ = 0$
 $m = \cos 60^\circ = \frac{1}{2}$
 $n = \cos 30^\circ = \frac{\sqrt{3}}{2}$

1] If a line makes $\angle$'s $90^\circ, 135^\circ, 45^\circ$ with x, y, z axis. Find DC's.

Sol] $l = \cos \alpha = \cos 90^\circ = 0$
 $m = \cos \beta = \cos 135^\circ = -\frac{1}{\sqrt{2}}$ $\cos(180-45) = -\cos 45^\circ$
 $n = \cos \gamma = \cos 45^\circ = \frac{1}{\sqrt{2}}$

Eg 2] If a line has DR's $(2, -1, -2)$. Find its DC's.

Sol] DC's are:
 Consider: $\sqrt{4+1+4} = 3$
 $\therefore$ DC's are: $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$

** Eg 3] Find DC's of line passing through $(-2, 4, -5), (1, 2, 3)$

Sol] $A \equiv (-2, 4, -5), B \equiv (1, 2, 3)$
 DR's $\equiv (x_2 - x_1), (y_2 - y_1), (z_2 - z_1) = 3, -2, 8$
 consider $\sqrt{9+4+64} = \sqrt{77}$
 $\therefore$ DC's are: $\frac{3}{\sqrt{77}}, -\frac{2}{\sqrt{77}}, \frac{8}{\sqrt{77}}$

** (2m) 2] Find DC's of a line which makes equal angles with the coordinate axis.

Sol] Gn: equal angles $\Rightarrow \alpha = \beta = \gamma$
 also $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$
 $\cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1$
 $3 \cos^2 \alpha = 1$
 $\cos^2 \alpha = \frac{1}{3}$ $\cos \alpha = \pm \frac{1}{\sqrt{3}}$

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∴ DC's are $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}$

3] If a line has DR's $-18, 12, -4$, What are its DC's?
Sol] Consider $\sqrt{(-18)^2 + (12)^2 + (-4)^2} = 22$
∴ DC's are $\frac{-18}{22}, \frac{12}{22}, \frac{-4}{22}$ or $\frac{-9}{11}, \frac{6}{11}, \frac{-2}{11}$

Eg5] ST the pt's $A(2, 3, -4), B(1, -2, 3), C(3, 8, -11)$ are collinear.
Sol] DR's of $\vec{AB} : 1-2, -2-3, 3+4$ i.e. $-1, -5, 7$
DR's of $\vec{BC} : 3-1, 8+2, -11-3$ i.e. $2, 10, -14$
Clearly : $\frac{2}{-1} = \frac{10}{-5} = \frac{-14}{7} = -2$
∴ DR's of $\vec{AB}$ and $\vec{BC}$ are proportional.
 $\vec{AB} \parallel \vec{BC}$ but pt. B is common
∴ A, B, C are collinear.

4] ST the pt's $A(2, 3, 4), B(-1, -2, 1), C(5, 8, 7)$ are collinear.
Sol] DR's of $\vec{AB} : -1-2, -2-3, 1-4$ i.e. $-3, -5, -3$
DR's of $\vec{BC} : 5+1, 8+2, 7-1$ i.e. $6, 10, 6$
Clearly $\frac{6}{-3} = \frac{10}{-5} = \frac{6}{-3} = -2$
∴ DR's of $\vec{AB}$ and $\vec{BC}$ are proportional
 $\vec{AB} \parallel \vec{BC}$ but pt. B is common
∴ A, B, C are collinear.

5] Find the DC's of sides of the triangle whose vertices are $(3, 5, -4), (-1, 1, 2)$ and $(-5, -5, -2)$
Sol] $\vec{AB} : \text{DR's} = -4, -4, 6 ; \sqrt{16+16+36} = \sqrt{68} = 2\sqrt{17}$
DC's of $\vec{AB}$ are : $\frac{-4}{2\sqrt{17}}, \frac{-4}{2\sqrt{17}}, \frac{6}{2\sqrt{17}} = \frac{-2}{\sqrt{17}}, \frac{-2}{\sqrt{17}}, \frac{3}{\sqrt{17}}$

$$\vec{BC} : \text{D.R.'s} \equiv -4, -6, -4 ; \sqrt{16+36+16} = \sqrt{68} = 2\sqrt{17}$$

$$\text{D.C.'s of } \vec{AB} \text{ are : } \frac{-4}{2\sqrt{17}}, \frac{-6}{2\sqrt{17}}, \frac{-4}{2\sqrt{17}} \equiv \frac{-2}{\sqrt{17}}, \frac{-3}{\sqrt{17}}, \frac{-2}{\sqrt{17}}$$

$$\vec{AC} : \text{D.R.'s} \equiv +8, +10, -2 ; \sqrt{64+100+4} = \sqrt{168}$$

$$\text{D.C.'s of } \vec{AC} \text{ are : } \frac{+8}{\sqrt{168}}, \frac{+10}{\sqrt{168}}, \frac{-2}{\sqrt{168}}$$

$$\equiv \frac{+8}{2\sqrt{42}}, \frac{+10}{2\sqrt{42}}, \frac{-2}{2\sqrt{42}} \equiv \frac{+4}{\sqrt{42}}, \frac{+5}{\sqrt{42}}, \frac{-1}{\sqrt{42}}$$

Equation of a line in space :

NOTE: A line is uniquely determined if

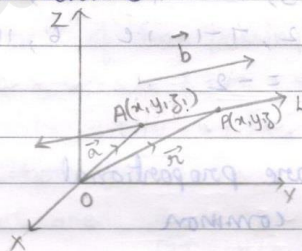
(a) it passes through a gn. pt. and has a gn. direction

$$(\vec{r} = \vec{a} + \lambda \vec{b})$$

pt. ||^{ve} vector

(b) it passes through two gn. pt.s $(\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}))$

I Equation of a line passing through a point and ||^{ve} to a given vector $\vec{b}$.



Let the line L pass through the gn. pt $A(x_1, y_1, z_1)$ with PV $\vec{a}$ and $\vec{b}$ be parallel to the gn. vector $\vec{b}$.

Let $P(x, y, z)$ be any arbitrary point on the line L with PV $\vec{r}$.

Now L is parallel to $\vec{b}$

$$\Rightarrow \vec{AP} \parallel \vec{b}$$

$$\Rightarrow \vec{AP} = \lambda \vec{b}$$

$$\vec{OP} - \vec{OA} = \lambda \vec{b}$$

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$\vec{r} - \vec{a} = \lambda \vec{b}$

Vector form: $\boxed{\vec{r} = \vec{a} + \lambda \vec{b}}$

Cartesian form: $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, $\vec{a} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$
Let $\vec{b} = a\vec{i} + b\vec{j} + c\vec{k}$

Substituting these in the vector form,

$$x\vec{i} + y\vec{j} + z\vec{k} = (x_1\vec{i} + y_1\vec{j} + z_1\vec{k}) + \lambda(a\vec{i} + b\vec{j} + c\vec{k})$$

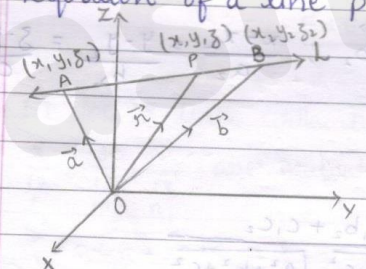
$$(x - x_1)\vec{i} + (y - y_1)\vec{j} + (z - z_1)\vec{k} = \lambda(a\vec{i} + b\vec{j} + c\vec{k})$$

Equating corresponding components,

$$\begin{array}{l|l|l} x - x_1 = \lambda a & y - y_1 = \lambda b & z - z_1 = \lambda c \\ \hline \frac{x - x_1}{a} = \lambda & \frac{y - y_1}{b} = \lambda & \frac{z - z_1}{c} = \lambda \end{array}$$

Cartesian form: $\boxed{\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}}$

II) Equation of a line passing through two points.



Let $\vec{a}$ and $\vec{b}$ be PV's of two given pts. A(x_1, y_1, z_1) and B(x_2, y_2, z_2) lying on a line L.

Let $\vec{r}$ be the PV of any arbitrary pt. P(x, y, z).

P is a point on the line if and only if $\vec{AP}$ and $\vec{AB}$ are collinear vectors.

$$\Rightarrow \vec{AP} \parallel \vec{AB}$$

$$\Rightarrow \vec{AP} = \lambda \vec{AB}$$

$$\vec{OP} - \vec{OA} = \lambda(\vec{OB} - \vec{OA})$$

$$\vec{r} - \vec{a} = \lambda(\vec{b} - \vec{a})$$

Vector form: $\boxed{\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})}$

Let $\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $\vec{a} = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$, $\vec{b} = a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$

Substituting these in vector form,

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k} = (x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}) + \lambda[(x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}]$$

$$(x - x_1)\mathbf{i} + (y - y_1)\mathbf{j} + (z - z_1)\mathbf{k} = \lambda[(x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}]$$

Equating the corresponding components

$$x - x_1 = \lambda(x_2 - x_1) \quad y - y_1 = \lambda(y_2 - y_1) \quad z - z_1 = \lambda(z_2 - z_1)$$

$$\frac{x - x_1}{x_2 - x_1} = \lambda \quad \frac{y - y_1}{y_2 - y_1} = \lambda \quad \frac{z - z_1}{z_2 - z_1} = \lambda$$

Eliminating the parameter λ ,

$$\text{Cartesian form: } \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Angle between two lines:

Formulae: $l_1: \vec{r} = \vec{a}_1 + \lambda\vec{b}_1$; $\frac{x - x_1}{a_1} = \frac{y - y_1}{b_1} = \frac{z - z_1}{c_1} \rightarrow \text{pts}$

$l_2: \vec{r} = \vec{a}_2 + \lambda\vec{b}_2$; $\frac{x - x_2}{a_2} = \frac{y - y_2}{b_2} = \frac{z - z_2}{c_2} \rightarrow \text{pts}$

Vector form: $\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$

Cartesian form: $\cos \theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

* If $\theta = 90^\circ$, the two lines are $\perp^e$

Condition is: $\vec{b}_1 \cdot \vec{b}_2 = 0 \Rightarrow l_1 \perp l_2$
(or) $a_1a_2 + b_1b_2 + c_1c_2 = 0 \Rightarrow l_1 \perp l_2$

* If $\theta = 0^\circ$, $l_1 \parallel l_2$

Conditions: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow l_1 \parallel l_2$

* Shortest distance between two skew lines:

vector form: $SD = d = \frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$

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Cartesian form: $d = \frac{|x_2 - x_1, y_2 - y_1, z_2 - z_1|}{\sqrt{(b_1c_2 - b_2c_1)^2 + (a_1c_2 - a_2c_1)^2 + (a_1b_2 - a_2b_1)^2}}$

* Distance between parallel lines: (Same $\parallel^e$ vectors) i.e. DR's

The lines are: $L_1: \vec{r} = \vec{a}_1 + \lambda \vec{b}$
 $L_2: \vec{r} = \vec{a}_2 + \mu \vec{b}$

Cartesian form: $L_1: \frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
 $L_2: \frac{x-x_2}{a} = \frac{y-y_2}{b} = \frac{z-z_2}{c}$

$d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$

EXERCISE 11.2

1] ST the 3 lines with DR's $\frac{12}{13}, \frac{-3}{13}, \frac{-4}{13}$; $\frac{4}{13}, \frac{12}{13}, \frac{3}{13}$; $\frac{3}{13}, \frac{-4}{13}, \frac{12}{13}$ are mutually $\perp^e$.

Sol.] Consider L_1 and L_2 : $d_1d_2 + m_1m_2 + n_1n_2 = \frac{48 - 36 - 12}{169} = 0$
 $\Rightarrow L_1 \perp^e L_2 \rightarrow \textcircled{1}$

Consider L_2 and L_3 : $d_2d_3 + m_2m_3 + n_2n_3 = \frac{12 - 48 + 36}{169} = 0$
 $\Rightarrow L_2 \perp^e L_3 \rightarrow \textcircled{2}$

Consider L_3 and L_1 : $d_3d_1 + m_3m_1 + n_3n_1 = \frac{36 + 12 - 48}{169} = 0$
 $L_3 \perp^e L_1 \rightarrow \textcircled{3}$

$\textcircled{1}, \textcircled{2}, \textcircled{3} \Rightarrow L_1, L_2, L_3$ are mutually $\perp^e$.

2] ST the line through the pts $(1, -1, 2), (3, 4, -2)$ is $\perp^{\text{ve}}$ to line through the pts $(0, 3, 2), (3, 5, 6)$.

Sol] L_1 : DR's are: $3-1, 4+1, -2-2 \equiv 2, 5, -4 \equiv a_1, b_1, c_1$

L_2 : DR's are: $3-0, 5-3, 6-2 \equiv 3, 2, 4 \equiv a_2, b_2, c_2$

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 6 + 10 - 16 = 0 \text{ and } a_1^2 + b_1^2 + c_1^2 \neq 0$$

$$\Rightarrow L_1 \perp L_2, \vec{d}_1 \cdot \vec{d}_2 = 0 \text{ and } |\vec{d}_1| \neq 0, |\vec{d}_2| \neq 0$$

3] ST the line through the pts $(4, 7, 8), (2, 3, 4)$ is $\parallel^{\text{ve}}$ to the line through the pts $(-1, -2, 1), (1, 2, 5)$.

Sol] L_1 : DR's are: $2-4, 3-7, 4-8 \equiv -2, -4, -4 \equiv a_1, b_1, c_1$

L_2 : DR's are: $1+1, 2+2, 5-1 \equiv 2, 4, 4 \equiv a_2, b_2, c_2$

$$\text{Clearly, } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = -1 \Rightarrow L_1 \parallel L_2$$

4] Find the eq. of the line which passes through the pt. $(1, 2, 3)$ and is $\parallel^{\text{ve}}$ to the vector $3\vec{i} + 2\vec{j} - 2\vec{k}$.

Sol] Gn: $A \equiv (1, 2, 3) \Rightarrow \vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}$

$\parallel^{\text{ve}}$ vector, $\vec{b} = 3\vec{i} + 2\vec{j} - 2\vec{k} \Rightarrow a=3, b=2, c=-2$

Vector form: $\vec{r} = \vec{a} + \lambda \vec{b}$

$$\vec{r} = (\vec{i} + 2\vec{j} + 3\vec{k}) + \lambda(3\vec{i} + 2\vec{j} - 2\vec{k})$$

$$\text{C-form: } \frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

$$\Rightarrow \frac{x-1}{3} = \frac{y-2}{2} = \frac{z-3}{-2}$$

5] Find eq. of line in vector and C-forms that passes through the pt. with pv $2\vec{i} - \vec{j} + 4\vec{k}$ and is in the direction of $\vec{i} + 2\vec{j} - \vec{k}$.

Sol] Gn: $A \equiv (2, -1, 4) \Rightarrow \vec{a} = 2\vec{i} - \vec{j} + 4\vec{k}$

$\parallel^{\text{ve}}$ vector, $\vec{b} = \vec{i} + 2\vec{j} - \vec{k} \Rightarrow a=1, b=2, c=-1$

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Vector form : $\vec{r} = \vec{a} + \lambda \vec{b}$
 $= (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + 2\hat{j} - \hat{k})$

C. form : $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
 $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-4}{-1}$

6] Find the C. eq. of the line passing through $(-2, 4, -5)$ and $\parallel$ to the line $\frac{x+3}{3} = \frac{y-4}{5} = \frac{z+8}{6}$

Sol] Gn : pt $\equiv (-2, 4, -5) \Rightarrow \vec{a} = -2\hat{i} + 4\hat{j} - 5\hat{k}$
 $\parallel$ vector to $\Rightarrow$ same DR's
 $\therefore \vec{b} = 3\hat{i} + 5\hat{j} + 6\hat{k}$ (using denominator of gn. eq.)
 $\Rightarrow a=3, b=5, c=6$

V. form : $\vec{r} = \vec{a} + \lambda \vec{b}$
 $= (-2\hat{i} + 4\hat{j} - 5\hat{k}) + \lambda(3\hat{i} + 5\hat{j} + 6\hat{k})$

C. eq : $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
 $\Rightarrow \frac{x+2}{3} = \frac{y-4}{5} = \frac{z+5}{6}$

7] The C. eq. of a line is $\frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$. Write its V. form

Sol] pt $\equiv A(5, -4, 6) \Rightarrow \vec{a} = 5\hat{i} - 4\hat{j} + 6\hat{k}$
DR's : $a=3, b=7, c=2 \Rightarrow \vec{b} = 3\hat{i} + 7\hat{j} + 2\hat{k}$
V. form : $\vec{r} = \vec{a} + \lambda \vec{b}$
 $= (5\hat{i} - 4\hat{j} + 6\hat{k}) + \lambda(3\hat{i} + 7\hat{j} + 2\hat{k})$

8] Find the vector and C. eq's of the line that passes through origin and $(5, -2, 3)$.

Sol] Gn: pts $A \equiv (0, 0, 0) \Rightarrow \vec{a} = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$

$B \equiv (5, -2, 3) \Rightarrow \vec{b} = 5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

$$\text{V-form: } \vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}) \\ = 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} + \lambda(5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k})$$

$$\text{C-form: } \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-0}{5-0} = \frac{y-0}{-2-0} = \frac{z-0}{3-0}$$

$$\Rightarrow \frac{x}{5} = \frac{y}{-2} = \frac{z}{3}$$

9] Find the vector and C.eq. of the line that passes through $(3, -2, -5)$ and $(3, -2, 6)$.

Sol] Gn: pts $A \equiv (3, -2, -5) \Rightarrow \vec{a} = 3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}$

$B \equiv (3, -2, 6) \Rightarrow \vec{b} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$

$$\text{V-form: } \vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}) \\ = (3\mathbf{i} - 2\mathbf{j} - 5\mathbf{k}) + \lambda(0\mathbf{i} + 0\mathbf{j} + 11\mathbf{k})$$

$$\text{C-form: } \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-3}{3-3} = \frac{y+2}{-2+2} = \frac{z+5}{6+5}$$

$$\frac{x-3}{0} = \frac{y+2}{0} = \frac{z+5}{11}$$

10] Find the $\angle$ b/w the following pairs of lines.

i] $\vec{r} = 2\mathbf{i} - 5\mathbf{j} + \mathbf{k} + \lambda(3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k})$ and

$\vec{r} = 7\mathbf{i} - 6\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} + 2\mathbf{k})$

Sol] $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1| |\vec{b}_2|}$

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$$\cos \theta = \frac{(3\hat{i} + 2\hat{j} + 6\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k})}{\sqrt{9 + 4 + 36} \sqrt{1 + 4 + 4}}$$

$$= \frac{3 + 4 + 12}{\sqrt{49} \sqrt{9}} = \frac{19}{7 \cdot 3} = \frac{19}{21}$$

$$\therefore \theta = \cos^{-1}\left(\frac{19}{21}\right)$$

ii] $\vec{r} = (3\hat{i} + \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} - 2\hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - 5\hat{k}) + \mu(3\hat{i} - \hat{j} - 4\hat{k})$

Sol] $\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1| |\vec{b}_2|} = \frac{3 + 5 + 8}{\sqrt{1+1+4} \sqrt{9+1+16}} = \frac{16}{\sqrt{6} \sqrt{50}} = \frac{16}{5\sqrt{2}\sqrt{3}\sqrt{6}}$

$$\cos \theta = \frac{16}{5 \cdot 2 \cdot \sqrt{3}} = \frac{8}{5\sqrt{3}}$$

$$\theta = \cos^{-1}\left(\frac{8}{5\sqrt{3}}\right)$$

11] Find the $\angle^{ie}$

i] $\frac{x-2}{2} = \frac{y-1}{5} = \frac{z+3}{-3}$ & $\frac{x+2}{1} = \frac{y-4}{8} = \frac{z-5}{4}$

Sol] $\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$

$$= \frac{-2 + 40 - 12}{\sqrt{4 + 25 + 9} \sqrt{1 + 64 + 16}} = \frac{26}{\sqrt{38} \sqrt{81}} = \frac{26}{9\sqrt{38}}$$

$$\theta = \cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$$

$$\text{ii)] } \frac{x}{2} = \frac{y}{2} = \frac{z}{1} \quad \& \quad \frac{x-5}{9} = \frac{y-2}{1} = \frac{z-3}{8}$$

$$\begin{aligned} \text{Sol)] } \cos \theta &= \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} \\ &= \frac{8 + 2 + 8}{\sqrt{4+4+1} \sqrt{16+1+64}} \\ &= \frac{18}{\sqrt{9} \sqrt{81}} = \frac{18}{3 \cdot 9} = \frac{18}{27} \end{aligned}$$

$$\therefore \theta = \cos^{-1}\left(\frac{18}{27}\right) = \cos^{-1}\left(\frac{2}{3}\right)$$

** 12] Find the values of p so that the lines $\frac{x-1}{3} = \frac{y-14}{2p} = \frac{z-3}{2}$ and $\frac{7-7x}{3p} = \frac{y-5}{1} = \frac{z-6}{5}$ are at $L^\perp$

$$\text{Sol)] } L_1: \frac{x-1}{-3} = \frac{y-14}{\frac{2p}{7}} = \frac{z-3}{2}, \quad L_2: \frac{x-1}{-\frac{3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5}$$

Since $L_1 \perp L_2$

$$\Rightarrow a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

$$\Rightarrow \frac{9p}{7} + \frac{2p}{7} - 10 = 0$$

$$\frac{11p}{7} - 10 = 0$$

$$11p = 70$$

$$p = \frac{70}{11}$$

13] ST the lines $\frac{x-5}{7} = \frac{y+2}{1} = \frac{z}{1}$ and $\frac{x}{2} = \frac{y}{3} = \frac{z}{3}$ are $L^\perp$.

$$\begin{aligned} \text{Sol)] } \text{Consider } a_1 a_2 + b_1 b_2 + c_1 c_2 \\ = 7 - 10 + 3 = 0 \end{aligned}$$

$$\therefore \vec{a} \cdot \vec{b} = 0$$

$\therefore$ The lines are $L^\perp$ to each other.

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13] Find the shortest distance b/w the lines $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$ and $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$

Sol] Gn: $L_1: \vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \lambda(7\hat{i} - 6\hat{j} + \hat{k})$
 $L_2: \vec{r} = (3\hat{i} + 5\hat{j} + 7\hat{k}) + \mu(\hat{i} - 2\hat{j} + \hat{k})$

$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

$\vec{a}_2 - \vec{a}_1 = (4\hat{i} + 6\hat{j} + 8\hat{k}) - (-\hat{i} - \hat{j} - \hat{k}) = 5\hat{i} + 7\hat{j} + 9\hat{k}$

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix} = \hat{i}(-6-2) - \hat{j}(7-1) + \hat{k}(-14-6) = -8\hat{i} - 6\hat{j} - 20\hat{k}$

$d = \frac{|(5\hat{i} + 7\hat{j} + 9\hat{k}) \cdot (-8\hat{i} - 6\hat{j} - 20\hat{k})|}{\sqrt{64 + 36 + 400}} = \frac{|-40 - 42 - 180|}{\sqrt{500}} = \frac{262}{\sqrt{500}} = \frac{131}{\sqrt{250}}$

14] Find the SD b/w the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$

Sol] $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

$\vec{a}_2 - \vec{a}_1 = (2\hat{i} - \hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k}) = \hat{i} - 3\hat{j} - 2\hat{k}$

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix} = \hat{i}(-2-1) - \hat{j}(2-2) + \hat{k}(1-2) = -3\hat{i} - \hat{k}$

$d = \frac{|(\hat{i} - 3\hat{j} - 2\hat{k}) \cdot (-3\hat{i} - \hat{k})|}{\sqrt{9 + 1}} = \frac{|-3 + 2|}{\sqrt{10}} = \frac{1}{\sqrt{10}}$

16] Find SD b/w the lines $\vec{r} = (\vec{a}_1 + 2\vec{j} + 3\vec{k}) + \lambda(\vec{i} - 3\vec{j} + 2\vec{k})$
 $\vec{r} = (4\vec{i} + 5\vec{j} + 6\vec{k}) + \mu(2\vec{i} + 3\vec{j} + \vec{k})$

Sol] $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

$\vec{a}_2 - \vec{a}_1 = 3\vec{i} + 3\vec{j} + 3\vec{k}$
 $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -3 & 2 \\ 2 & 3 & 1 \end{vmatrix}$
 $= \vec{i}(1 \cdot 1 - 2 \cdot 6) - \vec{j}(1 \cdot 1 - 4) + \vec{k}(3 + 6)$
 $= -11\vec{i} + 3\vec{j} + 9\vec{k}$

$d = \frac{|(3\vec{i} + 3\vec{j} + 3\vec{k}) \cdot (-11\vec{i} + 3\vec{j} + 9\vec{k})|}{\sqrt{121 + 9 + 81}}$
 $= \frac{|-33 + 9 + 27|}{\sqrt{211}}$
 $d = \frac{3}{\sqrt{19}}$ units

17] $\vec{r} = (1-t)\vec{i} + (t-2)\vec{j} + (3-2t)\vec{k}$ and $\vec{r} = (s+1)\vec{i} + (2s-1)\vec{j} + (2s+1)\vec{k}$

Sol] $L_1: \vec{r} = (\vec{a}_1 + 2\vec{j} + 3\vec{k}) + t(\vec{i} - 2\vec{j} - 2\vec{k})$
 $L_2: \vec{r} = (\vec{i} - \vec{j} - \vec{k}) + s(\vec{i} + 2\vec{j} - 2\vec{k})$

$\vec{a}_2 - \vec{a}_1 = \vec{j} - 4\vec{k}$
 $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -2 & -2 \\ 1 & 2 & -2 \end{vmatrix}$
 $= \vec{i}(4 - 4) - \vec{j}(-2 - 2) + \vec{k}(-2 - 2)$
 $= 4\vec{j} - 4\vec{k}$

$d = \frac{|(\vec{j} - 4\vec{k}) \cdot (4\vec{j} - 4\vec{k})|}{\sqrt{16 + 16}}$
 $= \frac{|4 - 16|}{\sqrt{32}}$
 $= \frac{12}{\sqrt{32}}$

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$$d = \frac{|-4 + 12|}{\sqrt{29}}$$

$$d = \frac{8}{\sqrt{29}} \text{ units}$$

Eg 11] Find SD b/w $\vec{r} = (i + j + 0k) + \lambda(2i - j + k)$ and $\vec{r} = (2i + j + (-1)k) + \mu(3i - 3j + 2k)$

Sol] $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$

$$\vec{a}_2 - \vec{a}_1 = i - k$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} i & j & k \\ 2 & -1 & 1 \\ 3 & -3 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} (i-k) \cdot (3i-j-7k) \\ \sqrt{9+1+9} \end{vmatrix}$$

$$= \frac{3+70}{\sqrt{59}} = \frac{73}{\sqrt{59}}$$

$$d = \frac{10}{\sqrt{59}} \text{ units}$$

Eg 12] Find distance b/w the lines L_1 and L_2 given by

$$\vec{r} = (i + 2j - 4k) + \lambda(2i + 3j + 6k) \text{ and } \vec{r} = (3i + 3j - 5k) + \mu(2i + 3j + 6k)$$

Sol] SD $d = \frac{|\vec{b}_1 \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$

$$\vec{a}_2 - \vec{a}_1 = 2i + j - k$$

$$\vec{b}_1 \times (\vec{a}_2 - \vec{a}_1) = \begin{vmatrix} i & j & k \\ 2 & 3 & 6 \\ 2 & 1 & -1 \end{vmatrix}$$

$$d = \frac{|-9i + 19j - 4k|}{\sqrt{4+9+36}}$$

$$= \frac{\sqrt{81+196+16}}{\sqrt{49}}$$

$$= \frac{\sqrt{293}}{7}$$

Equation of a Plane:

- * Eq. of a plane in normal form;

$$V. \text{ form : } \vec{r} \cdot \hat{n} = d \quad / \quad \vec{r} \cdot \vec{N} = d \quad \text{where } \frac{d}{|\vec{N}|} = b$$

$$C. \text{ form : } lx + my + nz = d \quad / \quad ax + by + cz = d$$

d = distance of the plane from origin.

- * Eq. of a plane $\perp$ to a given vector and passing through a gn pt.

$$V. \text{ form : } (\vec{r} - \vec{a}) \cdot \vec{N} = 0 \quad [(\vec{r} - \text{pt.}) \cdot \perp \text{ vector} = 0]$$

$$C. \text{ form : } A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$$

- * Eq. of a plane passing through 3 non-collinear pts. (STP)

$$V. \text{ form : } (\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$$

$$C. \text{ form : } \begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

- * Intercept form of the plane :

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

- (3m) * Eq. of the plane passing through intersection of two planes π_1 and π_2 is:

$$V. \text{ form : } \pi = \pi_1 + \lambda \pi_2 = 0$$

$$\text{Gn 2 planes : } C. \text{ form : } \pi_1 : A_1x + B_1y + C_1z + D_1 = 0$$

$$\pi_2 : A_2x + B_2y + C_2z + D_2 = 0$$

$$\text{New plane } \pi : (A_1x + B_1y + C_1z + D_1) + \lambda (A_2x + B_2y + C_2z + D_2)$$

- * Angle b/w two planes :

$$\pi_1 : \vec{r} \cdot \vec{N}_1 = d_1$$

$$\pi_2 : \vec{r} \cdot \vec{N}_2 = d_2$$

$$V. \text{ form : } \cos \theta = \frac{|\vec{N}_1 \cdot \vec{N}_2|}{|\vec{N}_1| |\vec{N}_2|}$$

$$C. \text{ form : } \cos \theta = \frac{|A_1A_2 + B_1B_2 + C_1C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}}$$

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**** Angle b/w a line and a plane :**

$L: \vec{r} = \vec{a} + \lambda \vec{b}$

$\Pi: \vec{r} \cdot \vec{n} = d$

$\sin \phi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|}$

(1m) * Distance of a pt. from a plane :

$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$

EXERCISE 11.3

14] In the following cases, find the distance of gn. pts from the plane.

(a) $pt \equiv (0, 0, 0); \Pi: 3x - 4y + 12z = 3 \Rightarrow 0 = 0$

$d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|0 + 0 + 0 - 3|}{\sqrt{9 + 16 + 144}}$

$d = \frac{|-3|}{\sqrt{169}} = \frac{3}{13} \text{ units}$

(b) $pt \equiv (3, -2, 1); \Pi: 2x - y + 2z + 3 = 0$

$d = \frac{|6 + 2 + 2 + 3|}{\sqrt{4 + 1 + 4}} = \frac{|13|}{\sqrt{9}} = \frac{13}{3} \text{ units}$

(c) $pt \equiv (2, 3, -5); \Pi: x + 2y - 2z = 9$

$d = \frac{|2 + 6 + 10 - 9|}{\sqrt{1 + 4 + 4}} = \frac{|9|}{\sqrt{9}} = \frac{9}{3} = 3 \text{ units}$

(d) $pt \equiv (-6, 0, 0); \Pi: 2x - 3y + 6z - 2 = 0$

$d = \frac{|-12 - 2|}{\sqrt{4 + 9 + 36}} = \frac{|-14|}{\sqrt{49}} = \frac{14}{7} = 2 \text{ units}$

13] In the following cases determine whether the given planes are $\parallel$ or $\perp$ and in case they are neither, find $\angle$ b/w them

NOTE: (a) $A_1A_2 + B_1B_2 + C_1C_2 = 0 \Rightarrow \Pi_1 \perp \Pi_2$

(b) $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2} = \lambda \Rightarrow \Pi_1 \parallel \Pi_2$

$$(c) \cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|}$$

(a) $7x + 5y + 6z + 30 = 0$ and $3x - y - 10z + 4 = 0$

Sol] Clearly Π_1 and Π_2 are neither $\parallel$ nor $\perp$.

$$\cos \theta = \frac{|A_1A_2 + B_1B_2 + C_1C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}}$$

$$= \frac{|21 - 5 - 60|}{\sqrt{49 + 25 + 36} \sqrt{9 + 1 + 100}} = \frac{-44}{\sqrt{110} \sqrt{110}} = \frac{-44}{110} = -\frac{2}{5}$$

$$\cos \theta = -\frac{2}{5}$$

$$\theta = \cos^{-1}\left(-\frac{2}{5}\right)$$

(b) $2x + y + 3z - 2 = 0$ and $x - 2y + 5 = 0$

Sol] Clearly Π_1 and Π_2 are neither $\parallel$ nor $\perp$.

$$\text{consider: } A_1A_2 + B_1B_2 + C_1C_2 = 2 + 5 + 0 = 7 \neq 0$$

$$= 2 - 2 + 0 = 0$$

$$\therefore \Pi_1 \perp \Pi_2$$

(c) $2x - 2y + 4z + 5 = 0$ and $3x - 3y + 6z - 1 = 0$

Sol] Consider $\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2} = \lambda \Rightarrow \frac{2}{3} = \frac{-2}{-3} = \frac{4}{6} = \frac{5}{-1} = \frac{2}{3}$

$$\therefore \Pi_1 \parallel \Pi_2$$

(d) $2x - y + 3z - 1 = 0$ and $2x - y + 3z + 3 = 0$

$$\text{Sol] } \frac{2}{2} = \frac{-1}{-1} = \frac{3}{3} = 1$$

$$\therefore \Pi_1 \parallel \Pi_2$$

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(e) $4x + 8y + z - 8 = 0$ and $y + z - 4 = 0$

Sol] $\cos \theta = \frac{|8 + 1|}{\sqrt{16 + 64 + 1} \sqrt{1 + 1}} = \frac{9}{\sqrt{81} \sqrt{2}} = \frac{1}{\sqrt{2}}$

$\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$

12] Find the angle b/w the planes whose vector eq.s are
 $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 5$ and $\vec{r} \cdot (3\hat{i} - 3\hat{j} + 5\hat{k}) = 3$

Sol] $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{|6 - 6 - 15|}{\sqrt{4 + 4 + 9} \sqrt{9 + 9 + 25}} = \frac{|-15|}{\sqrt{17} \sqrt{43}}$

$\theta = \cos^{-1}\left(\frac{15}{\sqrt{731}}\right)$

** 9] Find eq. of plane through the intersection of the planes
 $3x - y + 2z - 4 = 0$ and $x + y + z - 2 = 0$ and the pt $(2, 2, 1)$

Sol] Eq. of plane $= \pi_1 + \lambda \pi_2 = 0 \Rightarrow (x + y + z - 2) + \lambda (3x - y + 2z - 4) = 0$ (1)

(1) passes through $(2, 2, 1)$

$\Rightarrow (6 - 2 + 2 - 4) + \lambda (2 + 2 + 1 - 2) = 0$

$\Rightarrow 2 + \lambda 3 = 0$

$\lambda = -\frac{2}{3}$

(1) $\Rightarrow (3x - y + 2z - 4) - \frac{2}{3}(x + y + z - 2) = 0$

$(9x - 3y + 6z - 12) - 2x - 2y - 2z + 4 = 0$

$7x - 5y + 4z - 8 = 0$

10] Find the vector eq. of the plane passing through the
 intersection of $\vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 7$ and $\vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) = 9$
 and through the pt. $(2, 1, 3)$

Sol] Gn : $\Pi_1 : \vec{r} \cdot (2\hat{i} + 2\hat{j} - 3\hat{k}) = 7 \Rightarrow 2x + 2y - 3z - 7 = 0$ (1)

$\Pi_2 : \vec{r} \cdot (2\hat{i} + 5\hat{j} + 3\hat{k}) = 9 \Rightarrow 2x + 5y + 3z - 9 = 0$

Eq. of plane : $\Pi_1 + \lambda \Pi_2 = 0$

$(2x + 2y - 3z - 7) + \lambda(2x + 5y + 3z - 9) = 0 \rightarrow (1)$

① passes through $(2, 1, 3)$

$(4 + 2 - 9 - 7) + \lambda(4 + 5 + 9 - 9) = 0$

$(-10) + \lambda(9) = 0$

$\lambda = \frac{10}{9}$

① $\Rightarrow (2x + 2y - 3z - 7) + \frac{10}{9}(2x + 5y + 3z - 9) = 0$

$18x + 18y - 27z - 63 + 20x + 50y + 30z - 90 = 0$

$38x + 68y + 3z - 153 = 0$

V. form : $\vec{r} \cdot (38\hat{i} + 68\hat{j} + 3\hat{k}) = 153$

Ex 20] Find the vector eq. of the plane passing through intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$ and the pt $(1, 1, 1)$.

Sol] $\Pi_1 : \vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6 \Rightarrow x + y + z - 6 = 0$

$\Pi_2 : \vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5 \Rightarrow 2x + 3y + 4z + 5 = 0$

Eq. of plane : $\Pi_1 + \lambda \Pi_2 = 0$

$(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0 \rightarrow (1)$

① passes through $(1, 1, 1)$

$(3 - 6) + \lambda(2 + 3 + 4 + 5) = 0$

$\lambda = \left(\frac{3}{14}\right) = \frac{3}{14}$

① $\Rightarrow (x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$

$5x + 5y + 5z - 30 + 2x + 3y + 4z + 5 = 0$

$7x + 8y + 9z - 25 = 0$

① $\Rightarrow (x + y + z - 6) + \frac{3}{14}(2x + 3y + 4z + 5) = 0$

$14x + 14y + 14z - 84 + 6x + 9y + 12z + 15 = 0$

C. form : $20x + 23y + 26z - 69 = 0$

V. form : $\vec{r} \cdot (20\hat{i} + 23\hat{j} + 26\hat{k}) = 69$

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11] Find the eq. of the plane through the line of intersection of the planes $x+y+z=1$ and $2x+3y+4z=5$ which is $\perp^e$ to the plane $x-y+z=0$

Sol] $\pi_1 : x+y+z=1$
 $\pi_2 : 2x+3y+4z=5$
 Eq. of plane through intersection of π_1 and π_2 is
 $\pi_1 + \lambda \pi_2 = 0$
 $(x+y+z-1) + \lambda(2x+3y+4z-5) = 0 \rightarrow (1)$
 (1) is $\perp^e$ to $x-y+z=0$
 $(a_1a_2+b_1b_2+c_1c_2)=0$
 $1(1+2\lambda) - 1(1+3\lambda) + 1(1+4\lambda) = 0$
 $1+2\lambda - 1-3\lambda + 1+4\lambda = 0$
 $1+3\lambda = 0$
 $\lambda = -\frac{1}{3}$
 (1) $\Rightarrow (x+y+z-1) + (-\frac{1}{3})(2x+3y+4z-5) = 0$
 $3x+3y+3z-3 - 2x-3y-4z+5 = 0$
 $x-z+2=0$

12] (a) Find the eq. of the plane with intercept 3 on y-axis and parallel to xoz plane.

Sol] $y=3$

(b) Find eq. of plane making intercept of -2 on z-axis and parallel to xoy plane.

$z=-2 \Rightarrow z+2=0$

(c) Find eq. of plane making intercept of 5 on x-axis and $\parallel^e$ to yoz plane.

$x=5$

13] (a) Find the intercepts cut off by the plane $2x+y-z=5$

Sol] $\div$ by 5 : $\frac{2x}{5} + \frac{y}{5} - \frac{z}{5} = 1$
 $\Rightarrow x \text{ intercept} = a = \frac{5}{2}$ $z \text{ intercept} = c = -5$
 $y \text{ intercept} = b = 5$

(b) Find the eq. of plane making intercepts ~~1, 1, 1~~ $1, -1, 2$ on coordinate axis.

$$\frac{x}{1} + \frac{y}{-1} + \frac{z}{2} = 1$$

(c) Find the intercepts made by the plane $x - 3y + 6z = 6$

$$\frac{x}{6} + \frac{y}{-6/3} + \frac{z}{6/6} = 1 \Rightarrow \frac{x}{6} + \frac{y}{-2} + \frac{z}{1} = 1$$

$$\therefore x \text{ intercept} = a = 6$$

$$y \text{ " } = b = -2$$

$$z \text{ " } = c = 1$$

** 6] Find eq.s of the planes passing through the 3 pts

(a) $(1, 1, -1), (6, 4, -5), (-4, -2, 3)$

Eq. of plane through 3 non collinear pts.

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-1 & y-1 & z+1 \\ 6-1 & 4-1 & -5+1 \\ -4-1 & -2-1 & 3+1 \end{vmatrix} = 0$$

$$\begin{vmatrix} x-1 & y-1 & z+1 \\ 5 & 3 & -4 \\ -5 & -3 & 4 \end{vmatrix} = 0$$

$$R_2 = -R_3 \Rightarrow \Delta = 0$$

$\Rightarrow$ Gn. 3 pts are collinear

$\Rightarrow$ Infinite no. of planes pass through the gn. 3 pts.

(b) $(1, 1, 0), (1, 2, 1), (-2, 2, -1)$

$$\Rightarrow \begin{vmatrix} x-1 & y-1 & z-0 \\ 1-1 & 2-1 & 1-0 \\ -2-1 & 2-1 & -1-0 \end{vmatrix} = 0$$

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$$\begin{vmatrix} x-1 & y-1 & z-3 \\ 0 & 1 & 1 \\ -3 & 1 & -1 \end{vmatrix} = 0$$

$$x-1(-1-1) - (y-1)(3) + z(3) = 0$$

$$-2x + 2 - 3y + 3 + 3z = 0$$

$$2x + 3y - 3z - 5 = 0$$

NOTE: If $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$ → det'n of line L_1
 $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$ → det'n of line L_2

$$\Rightarrow \begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$\Rightarrow$ Gn lines are coplanar.

Eg 18] Find eq of plane passing through R(2, 5, -3), S(-2, -3, 5) and T(5, 3, -3) in vector and C-form.

Sol] V form: $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$

$\vec{a} = 2\hat{i} + 5\hat{j} - 3\hat{k}$, $\vec{b} = -2\hat{i} - 3\hat{j} + 5\hat{k}$, $\vec{c} = 5\hat{i} + 3\hat{j} - 3\hat{k}$

$\therefore [\vec{r} - (2\hat{i} + 5\hat{j} - 3\hat{k})] \cdot [(-4\hat{i} - 8\hat{j} + 8\hat{k}) \times (3\hat{i} - 2\hat{j} + 0\hat{k})] = 0$

C form: $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$

$$\begin{vmatrix} x-2 & y-5 & z+3 \\ -2-2 & -3-5 & 5+3 \\ 5-2 & 3-5 & -3+3 \end{vmatrix} = 0$$

$$\begin{vmatrix} x-2 & y-5 & z+3 \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = 0$$

$$(x-2)(16) - (y-5)(-24) + (z+3)(32) = 0$$

$$16x - 32 + 24y - 120 + 32z + 96 = 0$$

$$16x + 24y + 32z - 56 = 0$$

Eg 21] ST the lines $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$ and $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ are coplanar.

Sol] Here $(x_1, y_1, z_1) \equiv (-3, 1, 5)$

$(x_2, y_2, z_2) \equiv (-1, 2, 5)$

~~$a_1, b_1, c_1 \equiv$~~

$a_1, b_1, c_1 \equiv -3, 1, 5$ and $a_2, b_2, c_2 \equiv -1, 2, 5$

Consider

$x_2 - x_1$	$y_2 - y_1$	$z_2 - z_1$	a_1	a_2
a_1	b_1	c_1	b_2	c_2
a_2	b_2	c_2	a_1	a_2

$$\begin{vmatrix} 2 & 1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = 2(5-10) - 1(-15+5) = -10 + 10 = 0$$

$\therefore \Delta = 0 \Rightarrow$ Gn lines are coplanar.

Eg 22] Find the $\angle^{\circ}$ b/w the 2 planes $2x+y-2z=5$ and $3x-6y-2z=7$ using vector method.

Sol] Gn $\vec{n}_1 = 2\hat{i} + \hat{j} - 2\hat{k}$; $\vec{n}_2 = 3\hat{i} - 6\hat{j} - 2\hat{k}$

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{|6 - 6 + 4|}{\sqrt{4+1+4} \sqrt{9+36+4}}$$

$$\cos \theta = \frac{4}{\sqrt{9} \sqrt{49}} = \frac{4}{21}$$

$$\theta = \cos^{-1}\left(\frac{4}{21}\right)$$

Eg 23] Find the $\angle^{\circ}$ b/w 2 planes $3x-6y+2z=7$ and $2x+2y-2z=5$

$$\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} = \frac{|6 - 12 - 4|}{\sqrt{9+36+4} \sqrt{4+4+4}} = \frac{-10}{\sqrt{49} \sqrt{12}} = \frac{-10}{7\sqrt{3}}$$

$$\cos \theta = \frac{10}{7\sqrt{3}}$$

$$\theta = \cos^{-1}\left(\frac{10}{7\sqrt{3}}\right)$$



** Eg 24] Find the distance of a pt $(2, 5, -3)$ from the plane $\vec{r} \cdot (6\hat{i} - 3\hat{j} + 2\hat{k}) = 4$.

Sol] Gn: $\pi: 6x - 3y + 2z - 4 = 0$ pt $(2, 5, -3)$

$$d = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$= \frac{|12 - 15 - 6 - 4|}{\sqrt{36 + 9 + 4}} = \frac{13}{7} \text{ units}$$

** Eg 25] Find the $\angle^e$ b/w the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$

Sol] $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$; $\vec{n} = 10\hat{i} + 2\hat{j} - 11\hat{k}$

$$\sin \phi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}| |\vec{n}|} = \frac{|20 + 6 - 66|}{\sqrt{4+9+36} \sqrt{100+4+121}}$$

$$\sin \phi = \frac{|-40|}{7(15)} = \left(\frac{8}{21}\right)$$

$$\phi = \sin^{-1}\left(\frac{8}{21}\right)$$

III Shortest distance between skew lines:

Consider 2 skew lines $l_1: \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$, $l_2: \vec{r} = \vec{a}_2 + \lambda \vec{b}_2$

Let $\vec{a}_1$ and $\vec{a}_2$ be the position vectors of pt's S and T and $\vec{b}_1$ and $\vec{b}_2$ resp. of l_1 and l_2 resp. $\vec{PQ}$ is the shortest distance between l_1 and l_2 and is the projection of $\vec{ST}$ along $\vec{PQ}$.

$\vec{PQ}$ is $\perp$ to both $\vec{b}_1$ and $\vec{b}_2$, $\hat{n}$ being unit vector on $\vec{PQ}$.

$$\therefore \vec{PQ} = d\hat{n}; \hat{n} = \frac{\vec{b}_1 \times \vec{b}_2}{|\vec{b}_1 \times \vec{b}_2|}; |\vec{PQ}| = d$$

Let θ be the angle between $\vec{ST}$ and $\vec{PQ}$; $\vec{ST} = \vec{a}_2 - \vec{a}_1$

$$\cos \theta = \frac{\vec{ST} \cdot \vec{PQ}}{|\vec{ST}| |\vec{PQ}|}$$

$$0 = (\vec{n} \cdot (\vec{n} - \vec{r}_0))$$

$$\text{Also, } \cos \theta = \frac{|\vec{PQ}|}{|\vec{ST}|}$$

$$\Rightarrow \frac{|\vec{PQ}|}{|\vec{ST}|} = \frac{|\vec{ST} \cdot \vec{PQ}|}{|\vec{ST}| |\vec{PQ}|}$$

$$d = \frac{(\vec{a}_2 - \vec{a}_1) \cdot \vec{n}}{|\vec{n}|}$$

$$d = (\vec{a}_2 - \vec{a}_1) \cdot \hat{n}$$

$$\text{V. form: } d = \frac{(\vec{a}_2 - \vec{a}_1) \cdot \vec{b}_1 \times \vec{b}_2}{|\vec{b}_1 \times \vec{b}_2|}$$

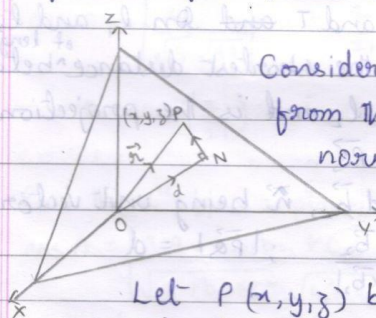
$$\text{In cartesian form: } l_1: \frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

$$l_2: \frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$$

Writing in determinant form, we get

$$d = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2-b_2c_1)^2 + (a_1c_2-a_2c_1)^2 + (a_1b_2-a_2b_1)^2}}$$

Equation of a plane in normal form:



Consider a plane whose $\perp^{\text{th}}$ distance from the origin is 'd'. If $\vec{ON}$ is the normal with $\hat{n}$ as unit normal vector, then $\vec{ON} = d\hat{n}$

Let $P(x, y, z)$ be any arbitrary pt. on the plane

Now, $\vec{NP} \perp \vec{ON}$

$$\Rightarrow \vec{NP} \cdot \vec{ON} = 0$$

$$(\vec{OP} - \vec{ON}) \cdot \vec{ON} = 0$$

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$$(\vec{r} - d\hat{n}) \cdot d\hat{n} = 0$$

$$(\vec{r} - d\hat{n}) \cdot \hat{n} = 0$$

$$\vec{r} \cdot \hat{n} - d\hat{n} \cdot \hat{n} = 0$$

$$\vec{r} \cdot \hat{n} - d = 0$$

V form: $\boxed{\vec{r} \cdot \hat{n} = d}$

Let $\vec{r} = xi + yj + zk$
 $\hat{n} = li + mj + nk$; (l, m, n being its D.C's)

Substituting these in vector form,

$$(xi + yj + zk) \cdot (li + mj + nk) = d$$

C form $\Rightarrow \boxed{lx + my + nz = d}$

5] Find the v & c eqs of the planes

(a) passes through $(1, 0, -2)$ and normal to the plane is $i + j - k$.

Sol] Gn: pt. $A \equiv (1, 0, -2) \Rightarrow \vec{a} = i - 2k$
 $\vec{N} = i + j - k \Rightarrow A = 1, B = 1, C = -1$

V form: $(\vec{r} - \vec{a}) \cdot \vec{N} = 0$
 $[\vec{r} - (i - 2k)] \cdot (i + j - k) = 0$

C form: $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$
 $1(x - 1) + 1(y - 0) - 1(z + 2) = 0$
 $x - 1 + y - z - 2 = 0$
 $\boxed{x + y - z - 3 = 0}$

(b) passing through $(1, 4, 6)$ and the normal vector to the plane is $i - 2j + k$.

Sol] Gn: pt. $A \equiv (1, 4, 6) \Rightarrow \vec{a} = i + 4j + 6k$
 $\vec{N} = i - 2j + k \Rightarrow A = 1, B = -2, C = 1$

V form: $(\vec{r} - \vec{a}) \cdot \vec{N} = 0$
 $[\vec{r} - (i + 4j + 6k)] \cdot (i - 2j + k) = 0$

$$\begin{aligned} \text{C-form: } A(x-x_1) + B(y-y_1) + C(z-z_1) &= 0 \\ 1(x-1) + -2(y-4) + 1(z-6) &= 0 \\ x-1-2y+8+z-6 &= 0 \\ \boxed{x-2y+z+1=0} \end{aligned}$$

3] Find the C.eq of the following planes.

(a) $\vec{r} \cdot (\vec{i} + \vec{j} - \vec{k}) = 2$

C-form: $x + y - z - 2 = 0$

(b) $\vec{r} \cdot (2\vec{i} + 3\vec{j} - 4\vec{k}) = 1$

C-form: $2x + 3y - 4z - 1 = 0$

(c) $\vec{r} \cdot [(s-2t)\vec{i} + (3-t)\vec{j} + (2s+t)\vec{k}] = 15$

C-form: $(s-2t)x + (3-t)y + (2s+t)z - 15 = 0$

2] Find v.eq of a plane which is at a distance of 7 units from the origin and normal to the vector $3\vec{i} + 5\vec{j} - 6\vec{k}$

Sol] Gn: $\vec{n} = 3\vec{i} + 5\vec{j} - 6\vec{k} \Rightarrow \hat{n} = \frac{3\vec{i} + 5\vec{j} - 6\vec{k}}{\sqrt{9+25+36}} = \frac{3\vec{i} + 5\vec{j} - 6\vec{k}}{\sqrt{70}}$

$d = 7$ units

v-form: $\vec{r} \cdot \hat{n} = d$

$\vec{r} \cdot \left(\frac{3\vec{i} + 5\vec{j} - 6\vec{k}}{\sqrt{70}} \right) = 7$

1] In each of the following cases, determine the DC's of the normal to the plane and the distance from the origin.

(b) $x + y + z = 1 \Rightarrow a=1, b=1, c=1$

DC's: $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}; d = \frac{1}{\sqrt{3}}$

(c) $2x + 3y - z = 5$

DC's: $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}; d = \frac{5}{\sqrt{14}}$

Gn. the gn. plane

$ax + by + cz = d$

$\Rightarrow a, b, c$ are DC's

$\therefore$ DC's are $\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$

1st dist. from origin = $\frac{d}{\sqrt{a^2+b^2+c^2}}$

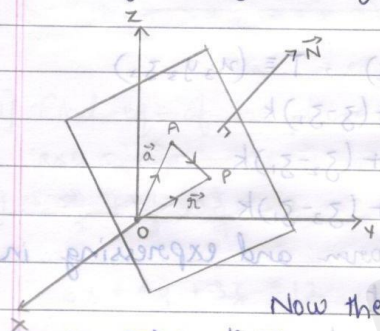
(a) $z = 2$

DC's: $0, 0, 1$; $d = 2$

(d) $5y + 8 = 0 \Rightarrow y = -\frac{8}{5}$

DC's: $0, 1, 0$; $d = \frac{+8}{5}$

V Equation of a plane perpendicular to given vector and passing through a given point.



Consider a plane perpendicular to given vector $\vec{N}$ and passing through the given pt. $A(x, y, z)$.

Let $P(x, y, z)$ be any arbitrary point in the plane with PV $\vec{r}$.

Now the pt. P lies on the plane if and only

if $\vec{AP}$ is $\perp$ to $\vec{N}$

$$\Rightarrow \vec{AP} \cdot \vec{N} = 0$$

$$(\vec{OP} - \vec{OA}) \cdot \vec{N} = 0$$

$$\text{V. form: } (\vec{r} - \vec{a}) \cdot \vec{N} = 0$$

$$\text{Let } \vec{r} = xi + yj + zk$$

$$\vec{a} = xi + yj + zk$$

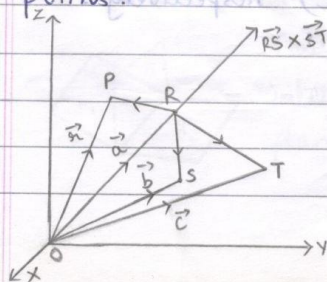
$$\vec{N} = Ai + Bj + Ck \text{ where } A, B, C \text{ are its DC's.}$$

Substituting these in the vector form,

$$[(x-x_1)i + (y-y_1)j + (z-z_1)k] \cdot (Ai + Bj + Ck) = 0$$

$$\Rightarrow A(x-x_1) + B(y-y_1) + C(z-z_1) = 0$$

VI Equation of a plane passing through 3 non-collinear points.



Let $\vec{a}, \vec{b}, \vec{c}$ be the PV's of the gn. 3 non-collinear pt's R, S, T resp.

Let $P(x, y, z)$ be any arbitrary pt. in the plane with PV $\vec{r}$.

The vectors $\vec{RS}$ and $\vec{RT}$ are in the gn. plane. $\therefore \vec{RS} \times \vec{RT}$ is $\perp^e$ to both the plane.

Eq. of the line plane passing through pt. R and $\perp^e$ to $\vec{RS} \times \vec{RT}$ is

$$(\vec{r} - \vec{a}) \cdot (\vec{RS} \times \vec{RT}) = 0$$

$$(\vec{r} - \vec{a}) \cdot [(\vec{OS} - \vec{OR}) \times (\vec{OT} - \vec{OR})] = 0$$

V. form: $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$

$$R \equiv (x_1, y_1, z_1), S \equiv (x_2, y_2, z_2), T \equiv (x_3, y_3, z_3)$$

$$\vec{RP} = (x - x_1)\vec{i} + (y - y_1)\vec{j} + (z - z_1)\vec{k}$$

$$\vec{RS} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j} + (z_2 - z_1)\vec{k}$$

$$\vec{RT} = (x_3 - x_1)\vec{i} + (y_3 - y_1)\vec{j} + (z_3 - z_1)\vec{k}$$

Substituting these in V. form and expressing in the determinant, we get

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$$

NOTE: If $\Delta = 0$, then the gn. 3 pts are collinear and infinite no. of planes pass through the gn. 3 collinear pts.

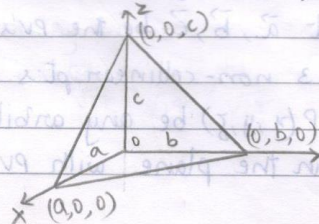
VII Intercept form of the equation of a plane.

Consider a plane $Ax + By + Cz + D = 0 \rightarrow (1)$

Let the plane make intercepts a, b, c on the coordinate axes respectively.

Then plane passes through the pts

$(a, 0, 0), (0, b, 0), (0, 0, c)$ respectively.



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$$\Rightarrow \begin{array}{l|l|l} Aa + D = 0 & Bb + D = 0 & Cc + D = 0 \\ A = -\frac{D}{a} & B = -\frac{D}{b} & C = -\frac{D}{c} \end{array}$$

Substituting these in eq. ①

$$-\frac{Dx}{a} - \frac{Dy}{b} - \frac{Dz}{c} = -D$$

$$\div \text{ by } -D$$

$$\boxed{\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1}$$

Eg 14] Find eq. of the plane with intercepts 2, 3, 4 on coordinate axis resp.

Sol] $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$

$$6x + 4y + 3z = 12$$

Eg 15] Find the distance of the plane $2x + 3y + 4z - 6 = 0$ from the origin.

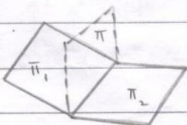
Sol] Consider $\sqrt{(2)^2 + (3)^2 + (4)^2} = \sqrt{4 + 9 + 16} = \sqrt{29}$

$\therefore$ DC's are $\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}}$

$\therefore$ distance from origin $= d = \frac{6}{\sqrt{29}}$

viii] Equation of a plane passing through the intersection of two planes.

Consider two planes π_1 and π_2 given by $\pi_1: \vec{r} \cdot \vec{n}_1 = d_1$ and $\pi_2: \vec{r} \cdot \vec{n}_2 = d_2$



Considering the P.V. on the line of intersection π_1 and π_2 , equation of the plane π is given by $\pi_1 + \lambda \pi_2 = 0$

$$(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$$

$$\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = (d_1 + \lambda d_2)$$

$$\text{V form: } \boxed{\vec{r} \cdot (\vec{n}_1 + \lambda \vec{n}_2) = (d_1 + \lambda d_2)}$$

$$\text{Let } \vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} ; \vec{n}_1 = A_1\mathbf{i} + B_1\mathbf{j} + C_1\mathbf{k} ; \vec{n}_2 = A_2\mathbf{i} + B_2\mathbf{j} + C_2\mathbf{k}$$

Substituting these in the vector form, we get

$$x(A_1 + \lambda A_2) + y(B_1 + \lambda B_2) + z(C_1 + \lambda C_2) = d_1 + \lambda d_2$$

$$\text{C form: } \boxed{(A_1x + B_1y + C_1z - d_1) + \lambda(A_2x + B_2y + C_2z - d_2) = 0}$$

MISCELLANEOUS EXERCISE:

- 1] ST the line joining the origin to the pt. ~~(2, 1, 1)~~ (2, 1, 1) is $\perp$ to the line determined by the pt's (3, 5, -1), (4, 3, -1)?

$$\text{Sol] } L_1: (0, 0, 0), (2, 1, 1) \Rightarrow \text{DR's: } 2, 1, 1$$

$$L_2: (3, 5, -1), (4, 3, -1) \Rightarrow \text{DR's: } 1, -2, 0$$

$$\text{Consider } a_1a_2 + b_1b_2 + c_1c_2 = 2 \cdot 1 + 1 \cdot (-2) + 1 \cdot 0 = 0$$

$$\Rightarrow L_1 \text{ is } \perp L_2$$

- 3] Find the angle b/w the lines whose DR's are a, b, c and $b-c, c-a, a-b$

$$\text{Sol] } L_1: \text{DR's: } a, b, c$$

$$L_2: \text{DR's: } b-c, c-a, a-b$$

$$\text{Consider } a_1a_2 + b_1b_2 + c_1c_2 = a(b-c) + b(c-a) + c(a-b)$$

$$= ab - ac + bc - ab + ac - bc = 0$$

$$\Rightarrow L_1 \text{ is } \perp L_2 \Rightarrow \theta = 90^\circ$$

4] Find the eq. of a line $\parallel$ to x-axis and passing through origin.

Sol] Giv: $\parallel^e$ vector = x-axis $\Rightarrow$ DR's : $\frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0}$

$$\text{pt} = (0, 0, 0)$$

Eq. of line : V-form : $\vec{r} = \vec{a} + \lambda \vec{b}$

$$\vec{r} = (0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k}) + \lambda(1\mathbf{i} + 0\mathbf{j} + 0\mathbf{k})$$

$$\text{C-form : } \frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0}$$

5] $A = (1, 2, 3)$, $B = (4, 5, 7)$, $C = (-4, 3, -6)$, $D = (2, 9, 2)$

Find the $\angle^e$ b/w $\vec{AB}$ and $\vec{CD}$

Sol] DR's of $\vec{AB}$: 3, 3, 4 DR's of $\vec{CD}$: 6, 6, 8

$$\text{Clearly } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$$

$\Rightarrow \vec{AB}$ is $\parallel^e$ $\vec{CD} \Rightarrow \theta = 0^\circ$

6] If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-1}{1} = \frac{z-6}{-5}$

are $\perp^e$, find the value of k .

Sol] Giv: $\perp^e$ $\Rightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$

$$(-3)(3k) + (2k)(1) + (2)(-5) = 0$$

$$-9k + 2k - 10 = 0$$

$$-7k - 10 = 0$$

$$-7k = 10$$

$$k = -\frac{10}{7}$$

** 7] Find the v.eq. of the line passing through $(1, 2, 3)$ and $\perp^e$

to the plane $\vec{r} \cdot (i + 2j - 5k) = 0$

Sol] $\vec{r} \cdot (i + 2j - 5k) = 0$

$\Rightarrow$ line is $\perp^e$ to plane $\Rightarrow$ same DR's

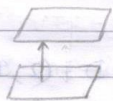
$$\therefore a = 1, b = 2, c = -5$$

For l: pt = (1, 2, 3) $\Rightarrow \vec{a} = i + 2j + 3k$
 $\vec{b} = i + 2j - 5k$

Eq. of l: v-form: $\vec{r} = \vec{a} + \lambda \vec{b}$
 $= (i + 2j + 3k) + \lambda(i + 2j - 5k)$

c-form: $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{-5}$

8] Find the eq. of the plane passing through pt (a, b, c) and $\parallel$ to the plane $\vec{r} \cdot (i + j + k) = 2$

Sol]  Let Gn: Π passes through (a, b, c)
 but Π is $\parallel$ to the gn. plane.
 $\therefore \Pi: \vec{r} \cdot (i + j + k) = p \rightarrow \text{①}$

① passes through (a, b, c)

$\therefore (ai + bj + ck) \cdot (i + j + k) = p$
 $a + b + c = p$

$\therefore \parallel \Pi: \boxed{\vec{r} \cdot (i + j + k) = a + b + c}$

9] Find the SD b/w the lines $\vec{r} = 6i + 2j + 2k + \lambda(i - 2j + 2k)$
 and $\vec{r} = -4i - k + \mu(3i - 2j - 2k)$

Sol] $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$
 $\vec{a}_2 - \vec{a}_1 = -10i - 2j - 3k$
 $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} i & j & k \\ 1 & -2 & 2 \\ 3 & -2 & -2 \end{vmatrix}$
 $= \frac{|(-10i - 2j - 3k) \cdot (8i + 8j + 4k)|}{\sqrt{64 + 64 + 16}}$
 $= \frac{|-80 - 16 - 12|}{12} = \frac{108}{12} = 9 \text{ units}$
 $= i(8) - j(-8) + k(4)$
 $= 8i + 8j + 4k$

10] Find the coordinates of the pt. where the line through (5, 1, 6) and (3, 4, 1) crosses the yz plane.

Sol] Let the pt. be R dividing yz plane in the ratio $\lambda:1$

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yz plane $\Rightarrow x=0$; pt $\equiv (5, 1, 6), (3, 4, 1)$
 $\Rightarrow$ pt is : $\left(\frac{5\lambda+3}{\lambda+1}, \frac{\lambda+4}{\lambda+1}, \frac{6\lambda+1}{\lambda+1} \right)$ section formula

$x=0 \Rightarrow \frac{5\lambda+3}{\lambda+1} = 0$
 $5\lambda+3=0$
 $\lambda = -\frac{3}{5}$

$\therefore$ pt $\equiv \left(0, \frac{-\frac{3}{5}+4}{-\frac{3}{5}+1}, \frac{-\frac{18}{5}+1}{-\frac{3}{5}+1} \right) \equiv \left(0, \frac{17}{2}, -\frac{13}{2} \right)$

11.] Find the coordinates of the pt. where the line through $(5, 1, 6)$ and $(3, 4, 1)$ crosses zx plane.
 Sol.] $y=0 \Rightarrow \frac{\lambda+4}{\lambda+1} = 0$
 $\lambda = -4$

$\therefore$ pt $\equiv \left(\frac{-17}{-3}, 0, \frac{-23}{-3} \right) \equiv \left(\frac{17}{3}, 0, \frac{23}{3} \right)$

Eg 30] Find the coordinates of the pt. where the line through $(5, 1, 6)$ and $(3, 4, 1)$ crosses xy plane.
 Sol.] $z=0 \Rightarrow \frac{6\lambda+1}{\lambda+1} = 0$
 $\lambda = -\frac{1}{6}$

$\therefore$ pt $\equiv \left(\frac{-\frac{5}{6}+3}{-\frac{1}{6}+1}, \frac{-\frac{6}{6}+4}{-\frac{1}{6}+1}, \frac{-1+1}{-\frac{1}{6}+1} \right) \equiv \left(\frac{13}{5}, \frac{23}{5}, 0 \right)$